University of Virginia Library Interlibrary Loan

Electronic Delivery Cover Sheet

WARNING CONCERNING COPYRIGHT RESTRICTIONS

The copyright law of the United States (Title 17, United States Code) governs the making of photocopies or other reproductions of copyrighted materials.

Under certain conditions specified in the law, libraries and archives are authorized to furnish a photocopy or other reproduction. One of these specified conditions is that the photocopy or reproduction is not to be "used for any purpose other than private study, scholarship, or research". If a user makes a request for, or later uses, a photocopy or reproduction for purposes in excess of "fair use", that user may be liable for copyright infringement.

This institution reserves the right to refuse to accept a copying order if, in its judgment, fulfillment of the order would involve violation of copyright law.

Interlibrary Loan (ILL) is for individual research, scholarship, and personal study only. Under U.S. copyright law (Title 17, Sections 107 & 108), ILL materials (items not owned by UVA Library) cannot replace course materials or required readings*. Faculty and instructors seeking to provide course content not within the library's collection should work with your department Subject Liaison Librarian to explore appropriate alternatives. For more on Fair Use, see the Instructional Sharing page.*

Intermetallics 18 (2010) 1586–1596

Contents lists available at ScienceDirect

Intermetallics

journal homepage: www.elsevier.com/locate/intermet

Influence of Y addition on the long time oxidation behaviors of high Nb containing TiAl alloys at 900 °C

L.L. Zhao, G.Y. Li, L.Q. Zhang, J.P. Lin*, X.P. Song, F. Ye, G.L. Chen

State Key Laboratory for Advanced Metals and Materials, University of Science and Technology Beijing, Beijing 100083, PR China

ARTICLE INFO

Article history:
Received 21 January 2010
Received in revised form
17 April 2010
Accepted 19 April 2010
Available online 10 May 2010

Keywords:
A. Titanium aluminides, based on TiAl
B. Microalloying
B. Oxidation
D. Microstructure

ABSTRACT

Influence of Y addition on the long time isothermal and cyclic oxidation behaviors of Ti45Al8Nb alloys containing 0.1–1.0 at.% Y with full lamellar microstructure was investigated in air at 900 °C. Only a proper amount of Y addition (0.3 at.%) can improve the long time oxidation resistance under both isothermal and cyclic oxidation. Either excessive or deficient addition of Y was detrimental to the oxidation resistance. Under isothermal oxidation, much thinner oxide scale was formed on 0.3 at.% Y-added alloy than it on Y-free alloy. Meanwhile, Y addition can effectively suppress the outmost TiO_2 layer and enhance the protective Al_2O_3 layer on the sample surface. Under cyclic oxidation, different from the oxide scale formed on Y-free and Y-lean (0.1 at.%) alloys, the oxide scales on higher Y-added alloys were not typical layered structure and oxide pegs protruding into the substrate were formed at the oxide scale/substrate interface. As a result, no spallation was observed in the higher Y-added alloys even after 1000 cycles while spallation occurred on Y-free alloy after 280 cycles and 0.1 at.% Y added alloy after 460 cycles. However, high Y added alloys (>0.3 at.%) suffered severe internal oxidation. Therefore, the oxidation rate on high Y added alloys was significantly high.

© 2010 Elsevier Ltd. All rights reserved.

1. Introduction

The lightweight TiAl alloys exhibit a large number of outstanding properties, such as high melting point, low density, intrinsic strength retention, high stiffness and creep resistance at elevated temperatures [1–3]. There are several potential applications that have been identified for TiAl-based alloys in the aerospace, automotive and gas turbine industries [4]. However, the actual service of TiAl alloy has been hindered by their poor oxidation resistance above 700 °C. A great deal of research work has been undertaken to improve the oxidation behavior of these alloys by adding ternary and quaternary elements. In view of third element addition, niobium and several other elements have been reported to be effective in the oxidation resistance of TiAl in Ref. [5–8]. The high Nb containing TiAl alloys are expected to use between 800 °C and 900 °C.

Alloys with higher Nb contents (up to 10 at.%) show much smaller mass gain in the isothermal oxidation. Niobium is firstly supposed to function as a dopant element with a higher valence than titanium, which is expected to decrease the oxygen vacancy concentration owing to the electroneutrality in the oxide, and thus to suppress rutile growth [9]. The addition of Nb also has other effect on the oxidation behaviors, such as increasing the activity of Al, lowering the solubility of oxygen in the alloy and promoting the formation of TiN at the oxide scale/substrate interface, which impede the diffusion of titanium and oxygen ions [10–12]. However, even 8–10 at.% Nb contents are not sufficient to form perfect Al_2O_3 layer, resulting in the transformation of the initially protective oxide scale to less protective one after a certain period of cyclic exposure. And scale spallation occurs under longer cyclic exposure. Studies have been done on rare-earth elements such as Y and Ce to improve the oxide scale adherence [13–15]. Benntt et al. found that Y-ion implantation considerably improved the corrosion resistance of high temperature alloys, particularly oxide scale adherence. In previous studies, Wu et al. investigated the influence of Y addition on TiAl-based alloys under isothermal and cyclic exposure at 800 °C and reported the novel possibility of improving the oxidation resistance without decreasing the mechanical properties by Y-modification. Although the beneficial effects of addition of rare-earth elements on oxidation resistance of various alloy systems were known [16–19], there was no comprehensive understanding of the effect of a rare-earth element Y on the oxidation behavior of TiAl-based alloys, especially on high Nb containing TiAl alloys. Therefore, the aim of the present work was to evaluate the influence of Y addition on the long time isothermal and cyclic oxidation behavior of high Nb containing TiAl alloys at 900 °C in air. Special emphasis was focused on examining of the oxide scale structures under isothermal and cyclic oxidation to

* Corresponding author. Tel.: +86 10 62332192; fax: +86 10 62332205.
E-mail address: linjunpin@skl.ustb.edu.cn (J.P. Lin).

0966-9795/$ – see front matter © 2010 Elsevier Ltd. All rights reserved.
doi:10.1016/j.intermet.2010.04.012

understand the role of Y on the oxide scale growth mechanism and spallation resistance.

2. Experimental

The nominal chemical compositions of the experimental materials were Ti45Al8Nb *x*Y (at.%), where $x = 0$, 0.1, 0.3, 0.4, 0.6, 0.8, 1.0. The alloys were prepared by tungsten-arc melting in an argon atmosphere, using titanium granular sponge (99.9%), pure aluminum pig and niobium–aluminum alloy. Then the solidification of the melted alloys occurred in a copper chill. To minimize compositional segregation the ingots were melted four times. Moreover, in order to obtain the full lamellar structure the ingots were heat treated at 1320 °C (in the α-phase field) for 12 h in a furnace and then transferred to 900 °C for 30 min in another furnace followed by air cooling.

For isothermal oxidation testing, the specimens were cut from the ingots using an electro-discharge machine (EDM). Specimens with approximate dimensions of $10 \times 10 \times 1$ mm^3 were mechanically polished up to 1200 grit SiC paper, and subsequently cleaned ultrasonically for 15 min in acetone. Isothermal oxidation tests were performed at 900 °C in a box furnace in static laboratory air for 100 h. The specimens were removed from the furnace after a particular period to measure the changes of their mass. The specimens were cooled in air and weighed using a balance with a resolution of 0.01 mg. Oxidation rate was evaluated by fixed-time weight gain measurement. At least two samples were tested in identical condition in order to verify the reproducibility of the data.

Cyclic oxidation tests were carried out in a cyclic oxidation furnace with an auto moving beam. The specimens were weighed after oxidation for some cycles. One cycle is composed of oxidation for 1 h at 900 °C and 12 min at ambient temperature. The specimens were inserted into and removed from the hot zone of the furnace within a few seconds to guarantee a rapid heating and cooling. When spallation of the oxide scale occurred, the oxidation cycles were recorded.

The surface and cross-section of samples after oxidation were examined by means of scanning electron microscopy (SEM) combined with an energy dispersive X-ray spectroscopy (EDS). The phases formed in the scales were identified by Rigaku D/max-RB X-ray diffractometry (Cu Kα).

3. Results

3.1. Effect of Y on the microstructure

Optical microstructures of the alloys before oxidation are shown in Fig. 1 which exhibits an equiaxed fully lamellar microstructure that consists of alternately arranged $\alpha_2 + \gamma$ platelets. The grain size gradually decreases with increasing Y content from 0 to 1.0 at.%. As shown in the SEM images of Y-added alloys (Fig. 2a and b), most of the white Y-rich phase is distributed along the grain boundaries and some fine Y-rich particles within the grains are mainly ellipsoids in shape. Furthermore, with the increasing of Y content up to 1.0%, Y-rich phase aggregated along grain boundaries is gradually changed to closed networks. EDS results show that the white phase segregated along grain boundaries is rich in Y and Al (the atomic ratio of Y to Al is about 1:2) while the ellipsoids within the grains are Y_2O_3 (Table 1). X-ray diffraction spectra of the Ti45Al8Nb1.0Y alloy confirm that Y-rich phase segregated along grain boundaries is YAl_2 phase, as shown in Fig. 2c.

3.2. Isothermal and cyclic oxidation kinetics

The curves of isothermal oxidation kinetics of the present alloys performed at 900 °C are shown in Fig. 3a. In order to understand whether a simple kinetics law can apply, the curves can be fitted by equation model as:

$$\Delta M^n = k_n t \tag{1}$$

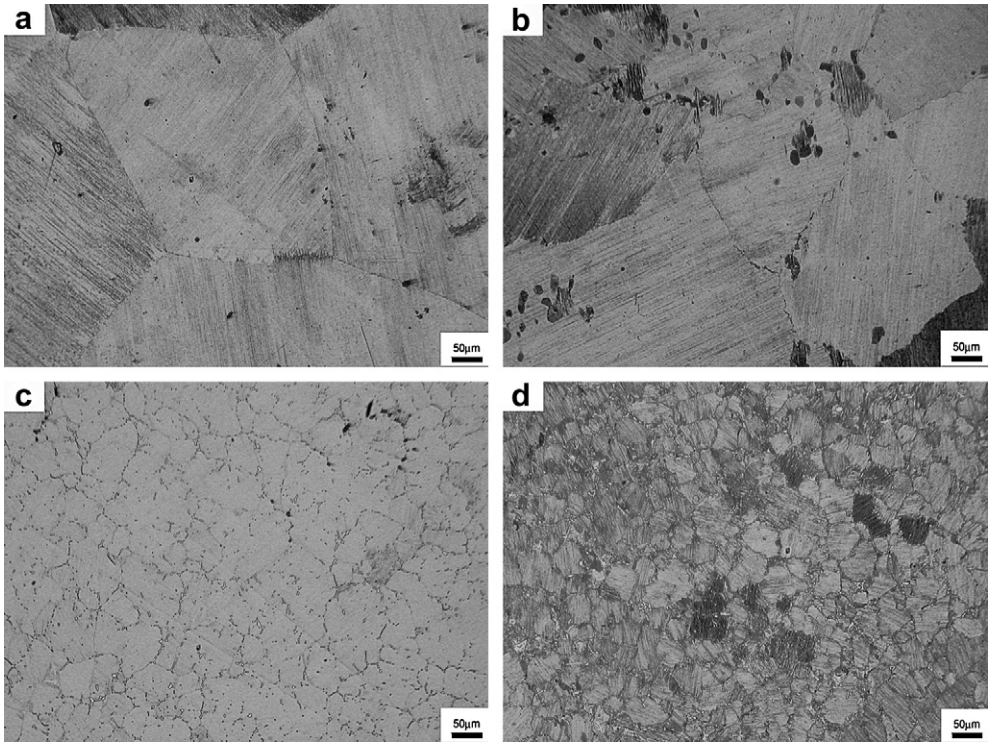


Fig. 1. Optical micrographs showing microstructure of Ti45Al8Nb alloys with different Y additions treated at 1320 °C/12 h/900 °C/30 min/AC: (a) Y-free, (b) 0.1 at.%Y, (c) 0.8 at.%Y and (d) 1.0 at.%Y. The grain size decreased with increasing the Y content.

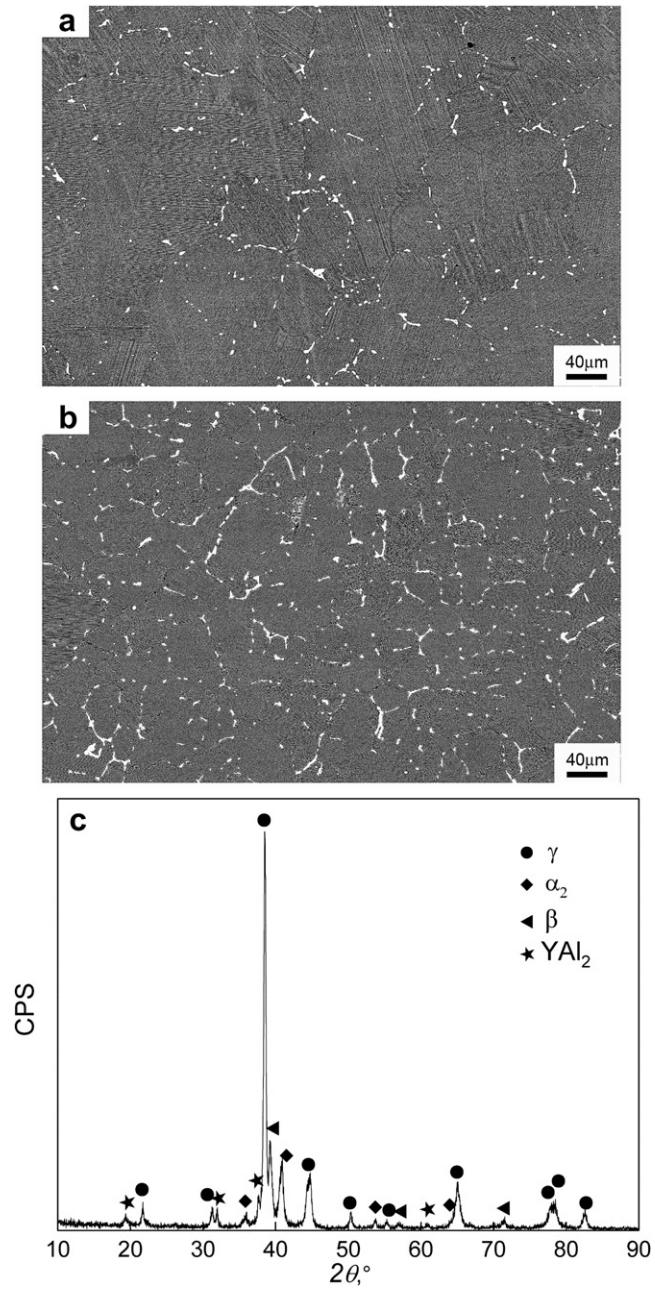


Fig. 2. SEM micrographs and XRD diffraction spectra of the Y-added alloys prior to oxidation: (a) 0.3 at.%Y, (b) 1.0 at.%Y and (c) 1.0 at.%Y.

where ΔM represents weight gain per unit area (mg/cm^2), n is the power exponent, k_n is the oxidation reaction rate constant (mg^n/cm^{2n}h) and t is the oxidation time (h). Thanks to the Origin software, it automatically tested successive values of n and k_n until the theoretic curve based on the Eq. (1) converged to the experimental

Table 1
EDS analysis of the white contrast phases in Fig. 2(a and b).

Element line	Grain boundaries		Within grains	
	Atom%	Atom% error	Atom%	Atom% error
Ti K	3.99	±0.09	1.08	±0.09
Al K	62.29	±0.31	1.92	±0.10
Y L	33.72	±0.17	38.59	±0.22
O K	0.0	–	58.40	±1.90

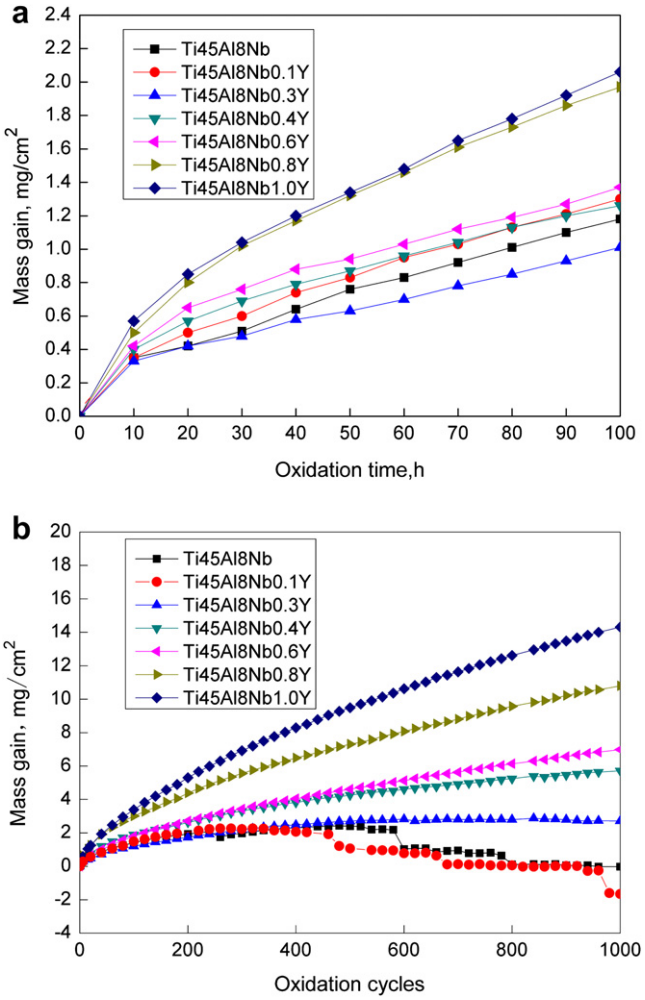


Fig. 3. Effect of Y on the oxidation kinetics of Ti45Al8Nb based alloys in air at 900 °C (a) under isothermal oxidation for 100 h and (b) under cyclic oxidation upon 1000 cycles. One cycle corresponds with 1 h at 900 °C and 12 min at ambient temperature.

mass gain curve. As Table 2 displays, the oxidation kinetic curves of all alloys obey a pseudo-parabolic or parabolic growth kinetics ($n \approx 2$) after early stage of oxidation (>20 h). In this case, the value of the oxidation reaction rate constant, k_n, would reflect the oxidation rates. The 0.3 at.% Y-added alloy has the lowest value of oxidation reaction rate constant, $k_n = 9.43 \times 10^{-3}$ mg^2/cm^4 h, meaning that it shows lowest oxidation rate. However, all other Y-added alloys have a larger k_n than that of Y-free alloy. The results indicate that either excessive or deficient addition of Y is detrimental to the oxidation resistance. Only a proper quantity of Y addition to the high-Nb TiAl alloy is beneficial to its long time oxidation resistance.

Cyclic oxidation tests performed at 900 °C demonstrated that Y-addition improved the oxide scale adherence to the substrate of

Table 2
Values of n and k_n of alloys isothermal oxidized at 900 °C for 100 h.

Alloys(at.%)	n	k_n(mg^n/cm^{2n} h)	t(h)
Ti45Al8Nb	1.6	0.01272	20–100
Ti45Al8Nb0.1Y	1.7	0.01522	20–100
Ti45Al8Nb0.3Y	1.9	0.00943	20–100
Ti45Al8Nb0.4Y	2.0	0.01572	0–100
Ti45Al8Nb0.6Y	2.1	0.01838	0–100
Ti45Al8Nb0.8Y	1.7	0.03285	0–100
Ti45Al8Nb1.0Y	1.8	0.03505	0–100

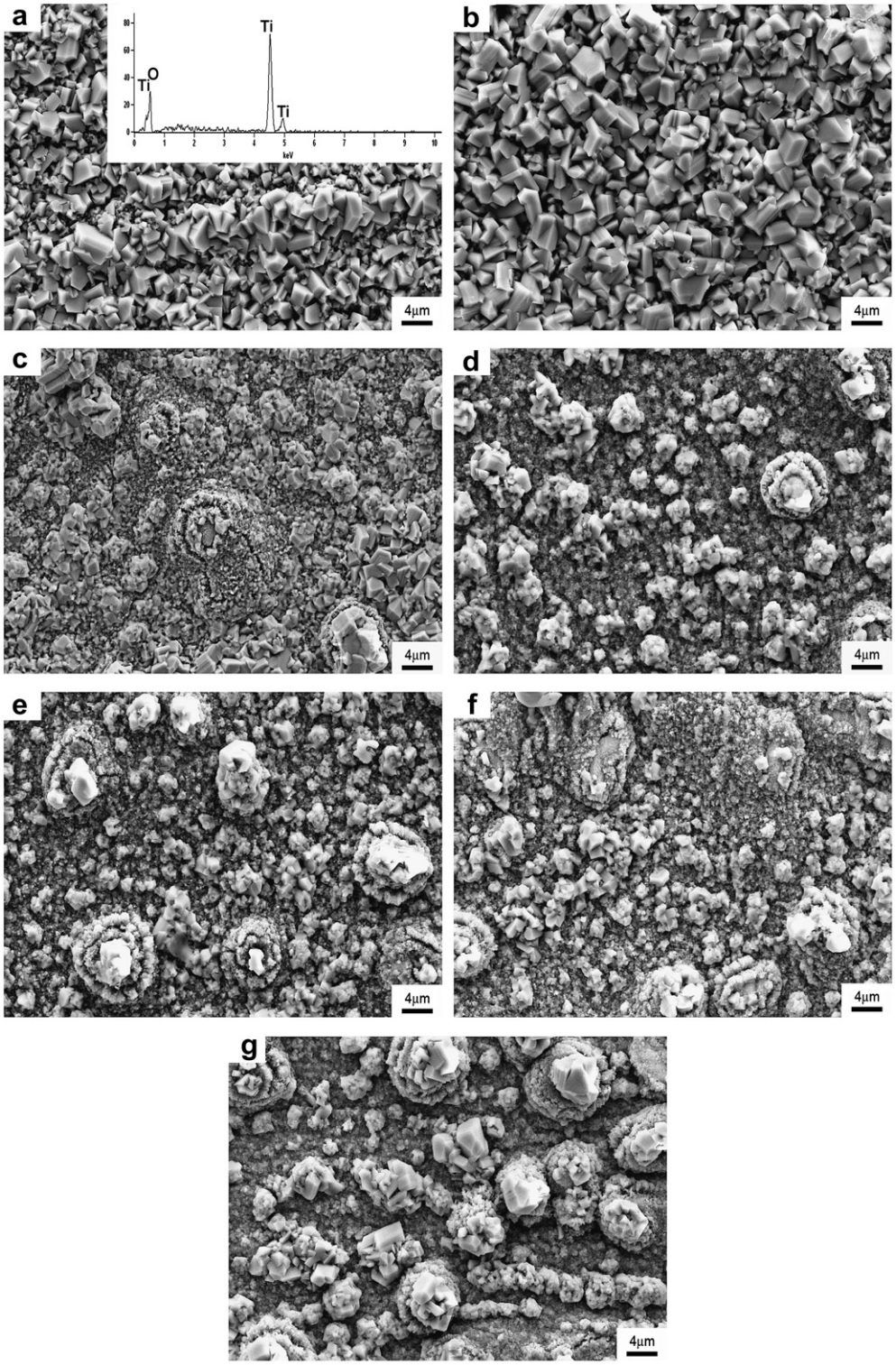


Fig. 4. SEM micrographs of the surface morphology of Ti45Al8Nb alloys with different Y additions after 100 h isothermal oxidation: (a) Y-free, (b) 0.1 at.%Y, (c) 0.3 at.%Y, (d) 0.4 at.%Y, (e) 0.6 at.%Y, (f) 0.8 at.%Y and (g) 1.0 at.%Y. Note: the inset is EDS spectra of the sample surface on Ti45Al8Nb alloy after 100 h isothermal oxidation.

Ti45Al8Nb based alloys. During the cyclic oxidation, mass loss is observed after hundreds of cycles in Y-free and 0.1 at.% Y-added alloy due to spallation as shown in Fig. 3b. The former alloy showed spallation after 280 oxidation cycles, while for the latter spallation appeared after 460 oxidation cycles. Besides, no spallation occurred on other Y-containing sample surfaces even after 1000 cycles. However, it is noteworthy that the mass gains of 0.4, 0.6, 0.8 and 1.0 at.% Y-added alloys are much higher than that of Ti45Al8Nb0.3Y alloy. Only the addition of 0.3 at.% Y can both improve the oxide scale adherence and decrease the oxidation rate.

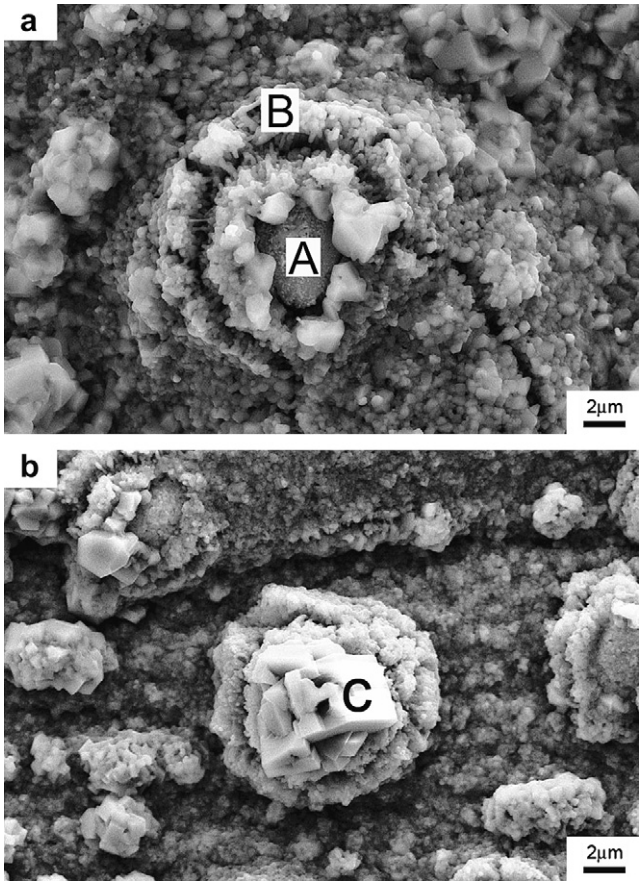


Fig. 5. SEM micrographs of the hillocks dispersed on the surfaces of higher Y-added alloys after 100 h isothermal oxidation: (a) 0.3 at.%Y and (b)1.01at.%Y.

3.3. Scale structures after isothermal oxidation

Fig. 4 shows the SEM images of the surface morphologies of the oxide scales for investigated alloys after 100 h isothermal oxidation. EDS spectra in the inset reveal that the surface of the Ti45Al8Nb alloy is covered by TiO_2 particles with different sizes. In our studies, we found that the TiO_2 particle showed a common feature of coarse-grained pillar-like crystal while the Al_2O_3 oxide was fine-grained particle without a regular shape. There is no difference in the surface morphology between 0.1 at.% Y-added alloy and Y-free alloy. However, when content of Y is higher than 0.1 at.% the surface morphologies are quite different from that of Y-free alloy. The feature of the surface morphologies of higher Y-added alloys is that there are some special hillocks dispersed on finer Al-rich oxide scales. The SEM micrographs in Fig. 5 show the details of the hillocks. The EDS results in Table 3 show that the hillock consists of coarse Y_2O_3 particle and fine Al_2O_3 grains in its surrounding. In addition, titanium oxides are formed on the top of the hillock due to limit solubility of Ti ions in both Y and Al oxides. More and more hillocks distribute on the surface with increasing content of Y. Besides the hillocks, the percent of coarse-grained pillar-like TiO_2 decreases with the increasing of Y addition on the sample surfaces after isothermal oxidation.

Table 3
EDS analysis of the selected regions of hillocks on Y-added sample surfaces shown in Fig. 5.

Element line	A		B		C	
	Atom%	Atom% error	Atom%	Atom% error	Atom%	Atom% error
O K	62.41	±1.94	72.37	±2.17	75.21	±6.61
Ti K	4.54	±0.14	3.90	±0.18	24.79	±0.87
Al K	6.19	±0.24	23.73	±0.41	0.0	–
Y L	26.86	±0.16	0.0	–	0.0	–

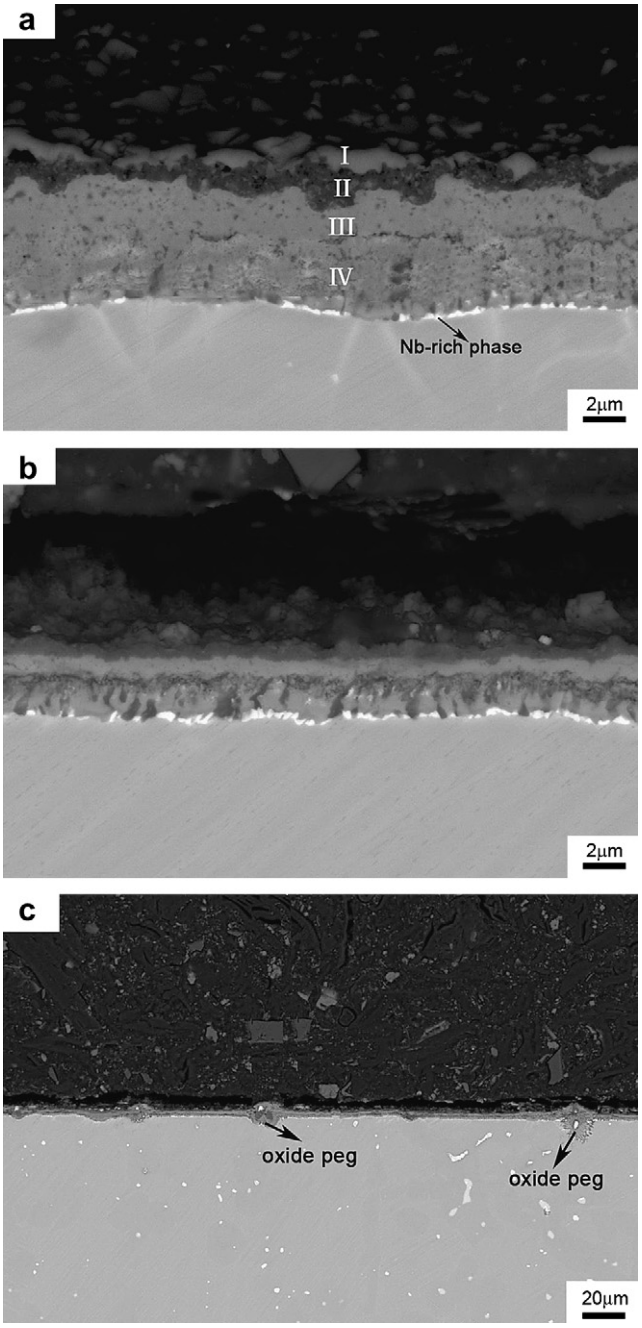


Fig. 6. SEM micrographs of the cross-sectional morphologies on (a) Ti45Al8Nb, (b) and (c) Ti45Al8Nb0.3Y alloys after 100 h isothermal oxidation. (c) is low-magnification SEM micrograph. Note: I–TiO_2 layer, II–Al_2O_3 layer, III–$(Ti,Nb)O_2$ –rich layer, IV–Mixed Al_2O_3 + $(Ti,Nb)O_2$ layer.

Fig. 6 shows SEM micrographs of the cross-sectional morphologies for Ti45Al8Nb and 0.3 at.% Y-added alloys after 100 h isothermal oxidation. In order to characterize the oxide scale structure, element distribution maps in cross-section and XRD analysis of the oxide scale are observed. Fig. 7 shows the element distribution maps of cross-section on Ti45Al8Nb alloy after 100 h isothermal oxidation shown in Fig. 6a. Fig. 8 shows the XRD

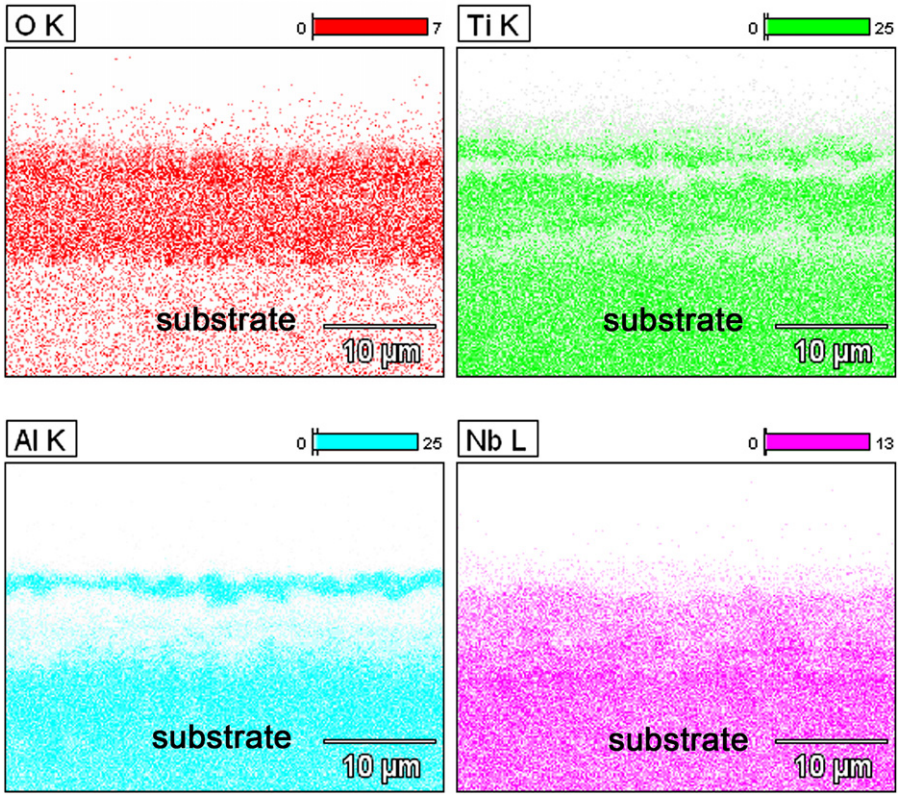


Fig. 7. Element distribution maps of the cross-section of Ti45Al8Nb alloy after 100 h isothermal oxidation shown in Fig. 6a.

diffraction spectra of the oxide scale formed on Ti45Al8Nb alloy after 100 h oxidation. The results show that gray contrast phase is TiO_2 and black contrast phase is Al_2O_3 in the BSE image of the cross-sectional morphologies (Fig. 6). In addition, no peaks of niobium oxide phase are detected in the XRD diffraction spectra of the oxide scale. In our previous work, Chen et al. found that Nb submitted for Ti to form $(Ti,Nb)O_2$ in the inner oxide layer [9]. Also the Nb-rich phase (white contrast in the BSE image) is observed at the oxide scale/substrate interface. We conclude that the oxide scale is composed of TiO_2 layer/Al_2O_3 layer/$(Ti,Nb)O_2$-rich layer/mixed $Al_2O_3 + (Ti,Nb)O_2$ oxide layer/Nb-rich transition layer from outside to inside. On 0.3 at.% Y-added alloy, Y can effectively reduce the thickness of the oxide scales (see Fig. 6b) and some oxide pegs protruding into the

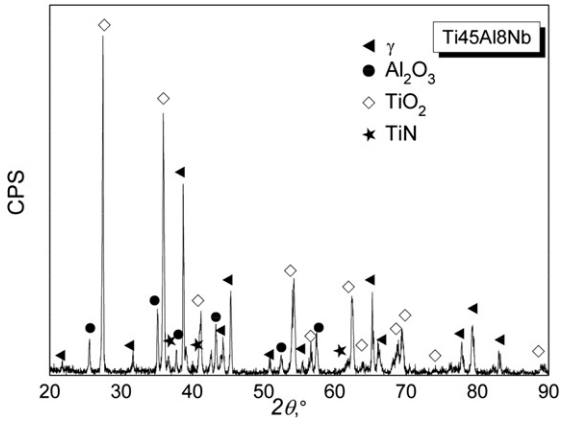


Fig. 8. XRD diffraction spectra of the oxide scale on Ti45Al8Nb alloy after 100 h isothermal oxidation.

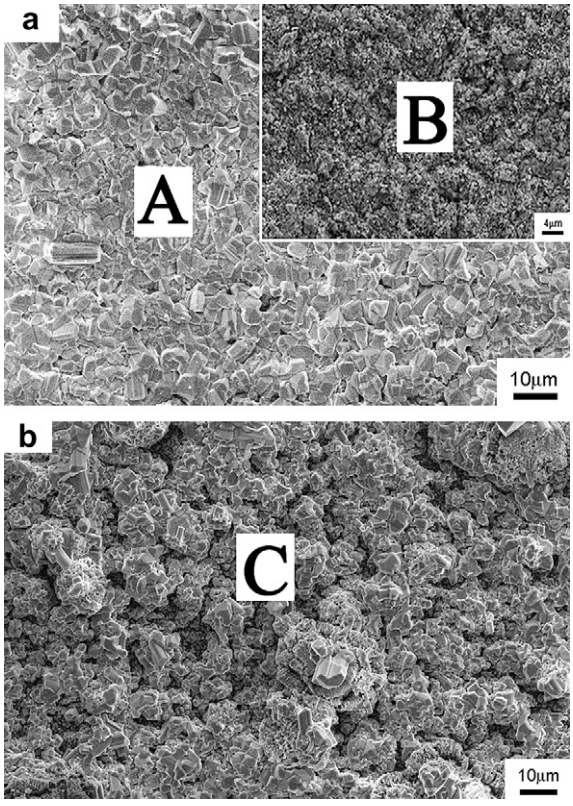


Fig. 9. SEM micrographs of the surface morphology of (a) Ti45Al8Nb alloy after 280 cycles and (b) 0.3 at.%Y added alloy after 1000 cycles on cyclic oxidation. Note: the inset is the SEM micrograph of the surface in spalled region.

Table 4
EDS analysis of the selected regions of the oxide scale surface after cyclic oxidation shown in Fig. 9.

Element line	A		B		C	
	Atom%	Atom% error	Atom%	Atom% error	Atom%	Atom% error
O K	62.29	±3.77	71.30	±3.21	66.96	±6.53
Ti K	28.13	±1.07	10.17	±0.70	16.82	±1.81
Al K	2.58	±0.24	15.23	±0.53	16.22	±1.08
Nb L	0.0	–	3.30	±0.20	0.0	–

Note: A—non-spalled area of the oxide scale on Y-free alloy upon 280 cycles, B—spalled area of the oxide scale on Y-free alloy upon 280 cycles oxidation and C—oxide scale surface on 0.3Y at.% added alloy upon 1000 cycles.

substrate can increase the contact areas of oxide scale and substrate, as shown in the low-magnification SEM micrograph (Fig. 6c). We also observe that no gray TiO_2 outmost layer but black Al_2O_3 outmost layer was formed on 0.3 at.% Y-added alloys.

3.4. Scale structures after cyclic oxidation

Fig. 9 shows the SEM micrographs of the surface morphologies for Ti45Al8Nb alloy after 280 cycles and 0.3 at.% Y-added alloys after 1000 cycles. After 280 cycles parts of the oxide scale on Ti45Al8Nb alloy have spalled off. EDS results (Table 4) reveal that non-spalled area of the oxide scale is covered with coarse TiO_2 crystals and spalled area is a finer grained structure rich in Ti, Al and O. Same results are found on the surface of Ti45Al8Nb0.1Y alloy after 460 cycles. On 0.3 at.% Y-added alloy no spallation occurred even after 1000 cycles. EDS results in Table 4 show that the surface is not covered with pure TiO_2 crystals but TiO_2 and Al_2O_3 mixed oxides. It is presumed that Y addition changed the mechanisms of the oxidation of high Nb containing TiAl alloys. Special hillocks also distribute on the surfaces after cyclic oxidation, as same as the surface morphologies for alloys after isothermal oxidation.

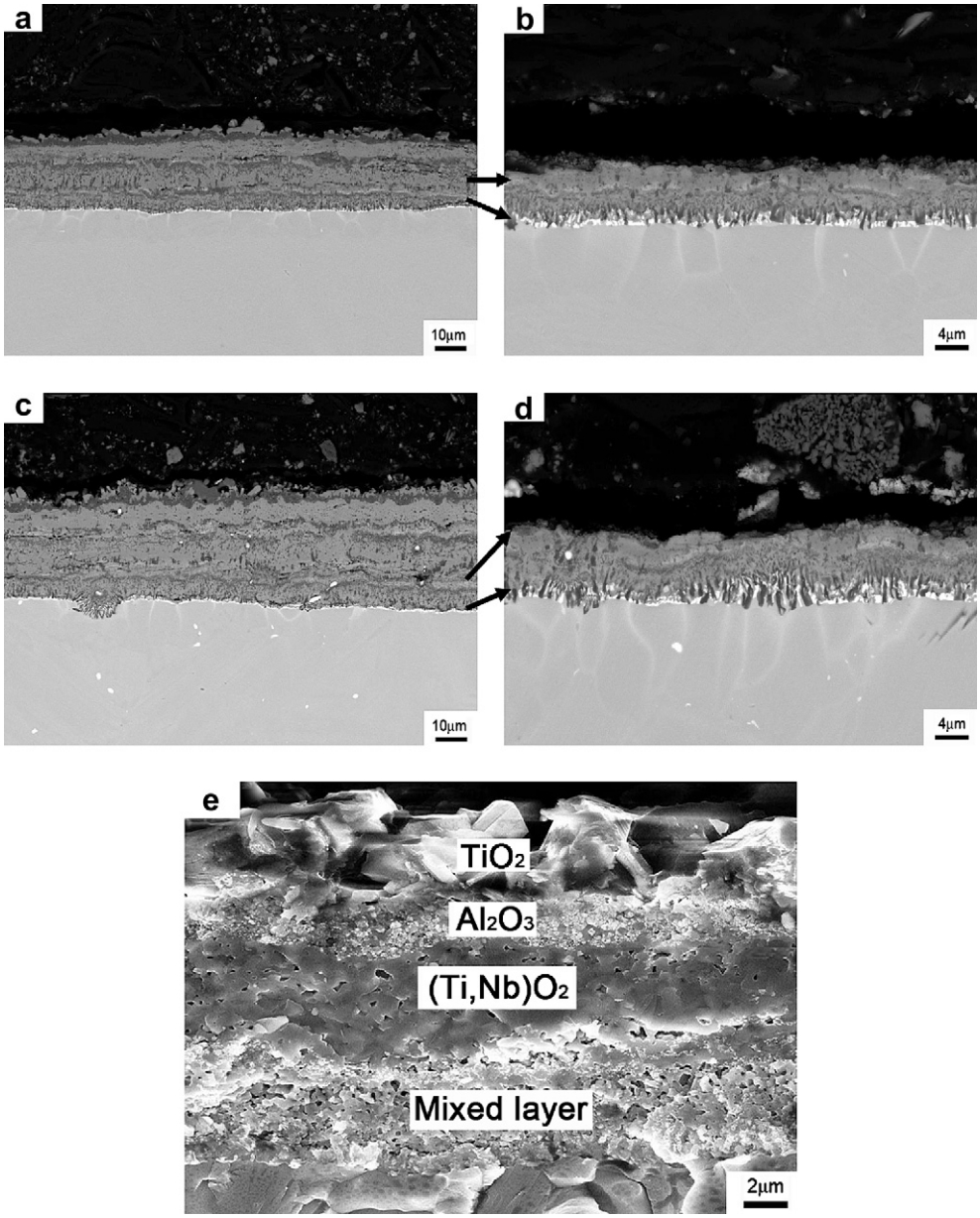


Fig. 10. SEM micrographs of the cross-sectional morphologies for Ti45Al8Nb alloy after 280 cycles (a,b),Ti45Al8Nb0.1Y alloy after 460 cycles (c,d) and high resolution secondary electron image of the cross-sectional morphology for Ti45Al8Nb alloy after 280 cycles(e) on cyclic oxidation. Note: cross-sectional morphologies after spalled in (b) and (d).

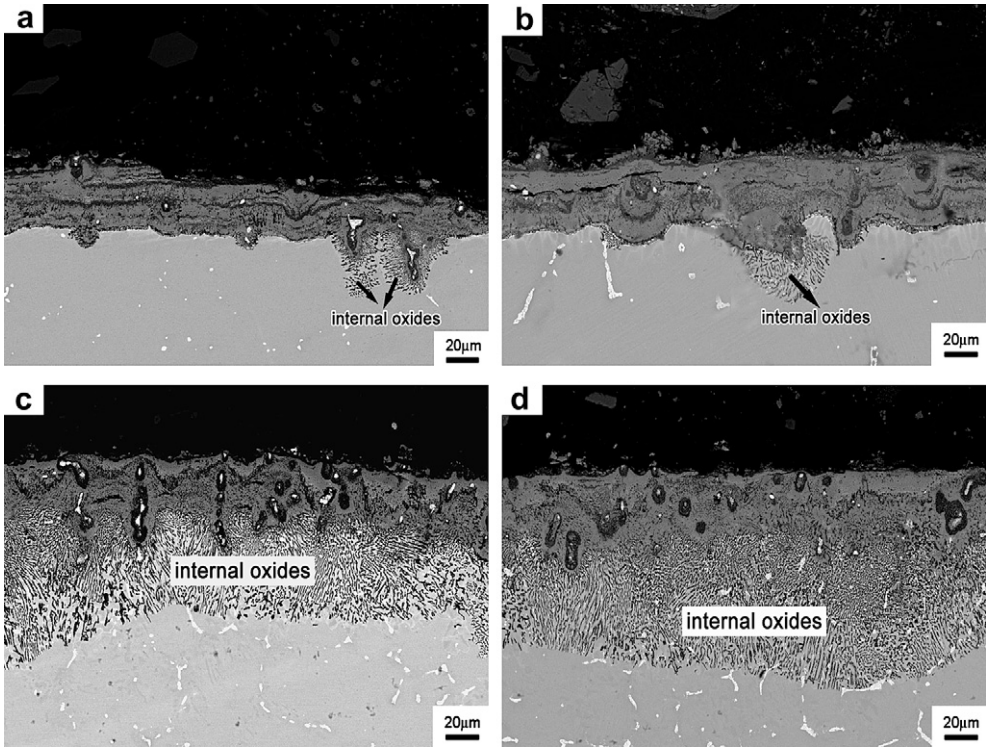


Fig. 11. SEM micrographs of the cross-sectional morphologies for Ti45Al8Nb alloys with different Y additions after 1000 cycles on cyclic oxidation: (a) 0.3 at.%Y, (b) 0.4Y at.%, (c) 0.8 at.%Y and (d) 1.0 at.%Y.

To further understand the structure of the oxide sale and the existence of Y, cross-sectional specimens were observed by means of SEM. Fig. 10 shows the micrographs of the cross-section of the oxide scale for Y-free alloy after 280 cycles and 0.1 at.% Y-added alloy after 460 cycles. The SEM micrographs of the cross-section on non-spalled area (Fig. 10a and c) reveal that the oxide scales are still typical multilayered structure and the inner mixed oxide layer is notably thickened after long time oxidation. As shown in Fig. 10b and d, only parts of inner mixed oxide layer are left after spallation. It can be concluded that cracking of the oxide scale occurred in the inner mixed oxide layer, which results in the spallation of outer oxide scale. The results are coincident with the EDS analysis (Table 4, point B) of the surface on spalled area, as shown in the inset in Fig. 9. Moreover, the high resolution SEM image of the cross-section of the scale formed in Y-free alloy after 280 cycles is shown in Fig. 10e. The Al_2O_3 layer has finer grains while the gain sizes of oxides in (Ti,Nb)O_2-rich layer are larger. However, the (Ti,Nb)O_2-rich oxides are much finer than TiO_2 outmost crystals. The outer layers are all compact and no cracks are found in them. Whereas, loose structure is observed in the mixed inner layer, which may be the reason for the detachment of oxide scale occurred in this porous layer during the cooling period.

Fig. 11 shows the micrographs of the cross-section of the oxide scale for 0.3, 0.6, 0.8 and 1.0 at.% Y-added alloys after 1000 h cyclic exposure at 900 °C. Different from the oxide scale formed on Y-free and Y-lean (0.1 at.%) alloys, the oxide scales on higher Y-added alloys are not typical layered structure and oxide pegs protruding into the substrate are formed at the oxide scale/substrate interface. Internal oxidation is detected on all Y-added alloys. On 0.3 at.% Y-added alloy, more continuous Al_2O_3 layers (black contrast) are formed and no thick mixed oxide layer exists in the oxide scale. With increasing of the Y content the extent of internal oxidation is more and more serious. Moreover, some white phase exists in the oxide scale on Y-added alloys. Element distribution maps in the cross-section of the oxide scale for 1.0 at.% Y-added alloy are shown in Fig. 12. Considering the results shown in Table 3, it is assumed that the white phase in the oxide scale is Y_2O_3 and black phase around it is Al_2O_3. The high resolution SEM image taken from the oxide scale on 0.8% Y-added alloy (Fig. 13) shows the morphologies of coarse-grained Y_2O_3 and fine-grained Al_2O_3 in its surrounding. This structure is formed by reaction of YAl_2 phase and oxygen at the grain boundaries. On high Y-added alloys YAl_2 phases aggregated along grain boundaries were gradually form closed networks (Fig. 2b). As a result, more and more Y_2O_3 were formed along the grain boundaries after exposure to 900 °C. The coarse-grained Y_2O_3 along the grain boundaries would supply the fast diffusion paths of the oxygen, aggravating the internal oxidation. The oxygen from air environment diffused along the grain boundaries could be further diffused into the lamellar grains along the phase boundaries between the γ/α_2 lamellae. As a result, black dendritic Al_2O_3 internal oxides in the inner layer formed on high Y-added alloy (see Fig. 11c and d).

4. Discussion

The present study demonstrates that a proper amount of Y is beneficial to oxidation resistance of high Nb containing TiAl alloys on both long time isothermal oxidation and cyclic oxidation. Generally, the solid solubility of Y in the α_2 and γ phases is less than 0.1 at.% [20]. When the content of Y is higher than 0.1 at.%, most of the Y elements accumulate along the grain boundaries and some fine particles are uniformly dispersed within the grains (Fig. 2). EDS and XRD results reveal that fine elliptical particles within the grains are Y_2O_3 while the white phases along the grain boundaries are YAl_2 compounds. Due to the strong binding between Y and O atoms, Y was prone to reacting with oxygen to form the Y_2O_3 during the melting processing of experimental alloys, which can reduce the amount of the interstitial O atoms. This effect of Y may be beneficial to the oxidation resistance of high Nb containing TiAl alloys.

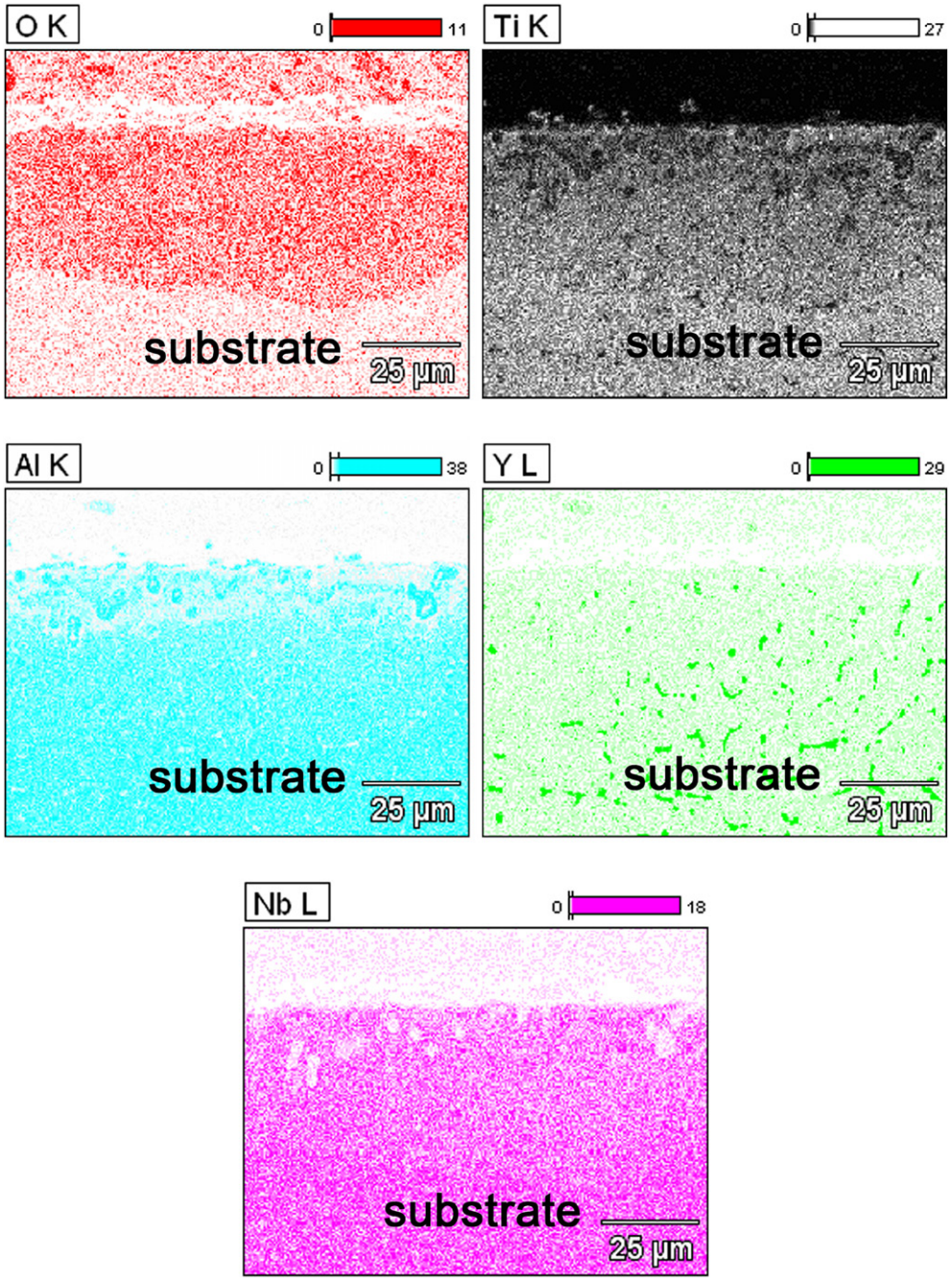


Fig. 12. Element distribution maps of the cross-section of Ti45Al8Nb1.0Y alloy after 1000 cycles on cyclic oxidation shown in Fig. 11d.

However, with increasing of Y content most of Y element aggregates along grain boundaries and gradually forms closed networks, which can reduce the grain sizes of the TiAl alloys. Li et al. reported that the large $YA1_2$ phases aggregated along the grain boundaries will harm the mechanical properties of high Y-containing TiAl alloys [21]. But it is still necessary to do further research on the effect of oxidation resistance.

Y addition suppressed the growth of TiO_2 at the surface of the samples during both isothermal oxidation and cyclic oxidation. It is inferred that Y addition changes the growth mechanisms of oxide scale on high Nb containing TiAl alloys. Our previous study has demonstrated that the oxidation of high Nb containing based alloys can be schematically divided into three stages [22]: (I)—selective growth of alumina-rich scale, (II)—growth of the Nb containing TiO_2 inner layer and (III)—formation of the outmost TiO_2 layer and inner mixed layer. Oxidation in stage I was controlled by the interface reaction while in stage II and stage III it was controlled by ions diffusion through oxide scale. The growth of the inner mixed TiO_2/Al_2O_3 layer was controlled by the inward diffusion of oxygen and the outer TiO_2 layer grew mainly by the outward diffusion of Ti ions. On Y-added alloys (higher than 0.1 at.%) YAl_2 segregated along the grain boundaries preferentially oxidized to Y_2O_3 and Al_2O_3 due to strong affinity of Y to oxygen. Thus, Y addition lowered the oxygen partial pressure within lamella colonies resulting in selective oxidation of Al. Protective Al_2O_3 outer layer is formed, as shown in Fig. 6b, and will be effective in retarding the ingress of the oxygen from the air environment after long time oxidation. To sum up, the oxidation in Y added alloy is controlled by inward diffusion of

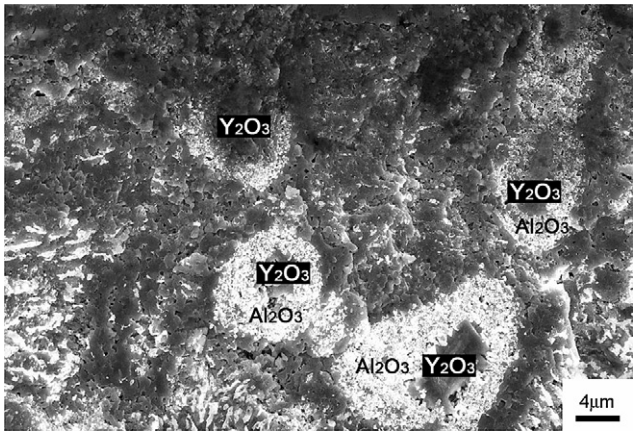


Fig. 13. The high resolution SEM image taken from the cross-section of oxide scale on 0.8% Y-added alloy after 1000 cycles on cyclic oxidation.

oxygen along grain boundaries and outward diffusion of Al within the lamellar colonies during long time oxidation. As a result, growth of the inner mixed TiO_2/Al_2O_3 layer and outmost TiO_2 layer are both suppressed on Y added alloys. From the analysis of the cross-sectional morphologies for Y-free and Y-lean (0.1 at.%) alloys after long time cyclic oxidation, as shown in Fig. 10, cracking of the oxide scale usually occurred in thick inner mixed layer during cooling period due to its loose structure. For this reason, Y addition can improve the resistance of the oxide scale spallation. In addition, oxides at the grain boundaries grow faster than it within grains. Therefore, oxides pegs protruding into the substrate are formed at the grain boundaries, which can increase the contact areas of oxide scale and substrate. It may be another reason of improvement to oxide scale adherence after long time oxidation on Y-added alloys. Strong refinement effect of Y results in a fine and closely packed oxide grain structure on Y added alloys. The fine grain layer structure guarantees more plasticity. Based on multi-analysis, no spallation of oxide scale would occur during cooling on Y-added alloy (>0.1 at.%) even after 1000 cycles on cyclic oxidation. All above described are the beneficial effects of Y addition on the scale spallation of the high Nb containing TiAl alloys.

However, coarse-grained Y_2O_3 and severe internal oxidation are observed on excessive Y-added alloys. They may have a detrimental effect on the oxidation resistance. The mass gains of high Y added alloys are much higher than the Y free alloy. Only 0.3 at.% Y-added alloy has lowest oxidation rate on both isothermal oxidation and cyclic oxidation. Excessive Y addition can cause considerable segregation of YAl_2 along the grain boundaries before oxidation (Fig. 2b). After exposure to 900 °C, severe intergranular corrosion occurred due to strong affinity of Y to oxygen at grain boundaries. Coarse grained Y_2O_3 along the grain boundaries would supply the fast diffusion paths of the oxygen resulting in severe internal oxidation after long time cyclic oxidation, as shown in Fig. 11b–d. The oxygen from air environment diffused along the grain boundaries could be further diffused into the lamellar colonies along the phase boundaries between the γ/α_2 lamellae. Then, the substrate was further oxidized and internal oxide scale was formed. Therefore, the oxidation rate of the high Y-added alloys was significantly high.

On the oxide scales of high Y-added alloys, dendritic Al_2O_3 internal oxides are formed in the inner layer. Regarding the dendritic Al_2O_3 internal oxides in inner oxide scale, the following possible reason is considered. According to the Vant Hoff equation, the Gibbs free energy change of oxidation can be defined as:

$$\Delta G = RT \ln \frac{p'_{O_2}}{p_{O_2}}$$

Where p'_{O_2} denotes the equilibrium oxygen partial pressure of oxidation reaction or dissociation pressure of oxide at reaction temperature, and p_{O_2} is the oxygen partial pressure of the oxidation environment. When $p_{O_2} > p'_{O_2}$, then $\Delta G < 0$, oxidation occurred. The paragraph of the $\Delta G^0 - T$ drawing by Richardson and Jeffes shows that p'_{O_2} of Al (10^{-36} atm) is smaller than it of Ti (10^{-30} atm). As a result, Al would preferentially oxidize at low oxygen partial pressure. The oxide scale would be a barrier of the ingress of oxygen from the air environment. Hence, the oxygen partial pressure is low beneath the oxide scale, which may be the reason of the formation of Al_2O_3 internal oxides.

Based on above analysis, 0.1 at.% Y addition does not have a significant effect on changing the growth mechanisms of oxide scale. The long time oxidation is also controlled by inward diffusion of O and outward diffusion of Ti as same as it on Y-free alloy. There are no differences between oxide scale on 0.1 at.% Y-added alloy and it on Y-free alloy, whether surface morphology or cross-sectional morphology. When Y content is higher than 0.1 at.%, no spallation occurs even after long time cyclic oxidation. However, excessive Y addition enhanced internal oxidation resulting in high oxidation rates. Only Ti45Al8Nb0.3Y alloy, in comparison with other alloys, shows highest isothermal and cyclic oxidation resistance after long time oxidation.

5. Conclusions

The morphological characteristics of oxide scales formed in Y-containing high Nb containing TiAl based alloys were studied under isothermal and cyclic oxidation, focusing on the existence of Y and the effect of Y on the oxidation resistance, and the following conclusions were drawn:

1. The addition of a proper amount of Y can improve the long time oxidation resistance under both isothermal and cyclic oxidation. A small amount of Y did not have a significant effect on the oxidation resistance. And excessive Y addition enhanced internal oxidation resulting in higher oxidation rates under both isothermal and cyclic oxidation.
2. Y addition changed the growth mechanisms of oxide scale. On 0.3 at.% Y-added alloy thinner oxide scale was formed under 100 h isothermal oxidation. Meanwhile, outmost TiO_2 layer was suppressed and protective Al_2O_3 layer was formed on the surface. Under cyclic oxidation, different from the oxide scale formed on Y-free and Y-lean (0.1 at.%) alloys, the oxide scales on higher Y-added alloys were not typical layered structure and oxide pegs protruding into the substrate were formed at the oxide scale/substrate interface.
3. No spallation of oxide scale was observed during cooling on Y-added alloys with Y content more than 0.1 at.% even after 1000 cycles on cyclic oxidation. However, coarse-grained Y_2O_3 and serve internal oxidation, which had a detrimental effect on the oxidation resistance, were observed on high Y-added alloy (>0.3 at.%). The coarse-grained Y_2O_3 segregated along the grain boundaries supplied fast diffusion path of oxygen anions. As a result, dendritic Al_2O_3 internal oxides were formed in the inner oxide scale. Therefore, the oxidation rate of high Y-added alloy was significantly high.

Acknowledgements

This work was financially supported by the National Natural Science Foundation of China (No.50771013 and No.50871127).

References

[1] Yamaguchi M, Inui H, Ito K. Acta Mater 2000;48:307–22.

[2] Liu CT. Mater Chem Phy 1995;42:77–86.
[3] Kim YW. Acta Metall Mater 1992;40:1121–34.
[4] Froes FH, Suryanarayana C, Eliezer D. ISIJ Int 1991;31:1235–48.
[5] Zhang YG, Li XY, Chen CQ, Wang W, Ji V. Surf Coat Technol 1998;100–101: 214–8.
[6] Stroosnijder MF, Zheng N, Quadakkers WJ, Hofman R, Gil A, Lanza F. Oxid Met 1996;46:19–35.
[7] Yoshihara M, Miura K. Intermetallics 1995;3:357–63.
[8] Shida Y, Anada H. Corros Sci 1993;35:945–53.
[9] Chen GL, Sun ZQ, Zhou X. Corros Sci 1992;48:939–46.
[10] Jiang H, Hirohasi M, Lu Y, Imanari H. Scr Mater 2002;46:639–43.
[11] Varma SK, Chan A, Mahapatra BN. Oxid Met 2001;55:423–35.
[12] Haanappel VA, Sunderkotter JD, Stroosnijder MF. Intermetallics 1999;7: 529–41.
[13] Wu Y, Umakoshi Y, Li X, Narita T. Oxid Met 2006;66:321–48.
[14] Wu Y, Hagihara K, Umakoshi Y. Intermetallics 2005;13:879–84.
[15] Stringer J. Mater Sci Eng A 1989;120–121:129–37.
[16] Cueff R, Buscail H, Caudron E, Issartel C, Riffard F. Corros Sci 2003;45: 1815–31.
[17] Bautista A, Velasco F, Abenojar J. Corros Sci 2003;45:1343–54.
[18] Thanneeru R, Patil S, Deshpande S, Seal S. Acta Mater 2007;55:3457–66.
[19] Xu CH, Gao W, Li S. Corros Sci 2001;43:671–88.
[20] Yang ZS, Zhang FZ, Ren LB, Zhou RZ, Yu ZS. Journal of University of Science and Technology Beijing; 1995. 17,p. 512–516.
[21] Li BH, Kong FT, Chen YY. J Rare Earth 2006;24:352–6.
[22] Zhao LL, Lin JP, Chen GL, Wang YL. In: Kim Y-W, Morris D, editors. Structural aluminides for elevated temperatures. Warrendale: TMS; 2008. p. 275–88.